

Day 04 Phase 6 â YouTube Transcript Connector

Objective

Implement a dedicated SpaceAgent YouTube connector route that supports metadata mode and transcript-backed mode when connector/runtime prerequisites exist.

Implementation Completed

New connector module

- 'src/lib/space-agent-youtube-connector.ts'
- parses YouTube URL/video id from request
- fetches metadata via YouTube oEmbed
- attempts transcript segment retrieval via 'youtube_transcript_api' only when runtime connector is available
- returns 'YouTubeResearchPacket' through existing packet contract

New API routes

- 'GET /api/gateway/space-agent/youtube/status'
- 'src/app/api/gateway/space-agent/youtube/status/route.ts'
- 'POST /api/gateway/space-agent/youtube/research'
- 'src/app/api/gateway/space-agent/youtube/research/route.ts'

New test coverage

- 'src/lib/space-agent-youtube-connector.test.ts'

Behavior Contract

1. Metadata mode always available if URL is valid and oEmbed returns data.
2. Transcript mode only when transcript connector runtime is available.
3. No full video download.
4. No bypass of login/age/region/paywall/copyright protections.
5. Truthful blocked reasons:
 - 'youtube_transcript_connector_not_proven'
 - 'youtube_transcript_unavailable'
 - 'youtube_transcript_lookup_failed'

Commands/Tests Run

- 'pnpm run typecheck' PASS
- 'pnpm run build' PASS
- 'pnpm run test' PASS

Proof

- Build route manifest includes:
 - '/api/gateway/space-agent/youtube/status'
 - '/api/gateway/space-agent/youtube/research'
- Connector test passes in full suite.

Current Live Status

- Connector path is implemented and test-proven in code.
- Transcript live proof is still environment-dependent on transcript runtime availability and authenticated endpoint execution.

Blocker (for transcript GO claim)

- 'youtube_transcript_connector_not_proven' (until runtime availability + live transcript smoke are confirmed in authorized environment).

Files Changed

- 'src/lib/space-agent-youtube-connector.ts'
- 'src/lib/space-agent-youtube-connector.test.ts'
- 'src/app/api/gateway/space-agent/youtube/status/route.ts'
- 'src/app/api/gateway/space-agent/youtube/research/route.ts'

Commits

- Pending Day 04 commit batch.

Rollback

- Revert files above to restore pre-connector limited route state.

No-Secrets Confirmation

- No secrets printed.
- No raw paths exposed in connector responses.
- No '.env' changes.

Updated Percentage

- YouTube track improved from vague LIMITED to explicit connector implementation + packet contract.
- Not marked GO until transcript smoke succeeds in live authorized run.

Exact Next Step

- Run one public transcript-enabled video request through the new research route in authorized environment and capture packet/timestamp evidence.